THE DATA ANALYST TOOLKIT

Excel · SQL · Power BI

How Excel powers analysis, how data pipelines feed it, and how the three core tools compare — topic by topic.

TOPIC 1 — TOPIC 2 — TOPIC 3 — TOPIC 4 — TOPIC 5

How Excel Is Used in Data Analysis?

Process	How Excel Is Used
Data Collection	Enter or import data from files, surveys, or other sources
Data Cleaning	Remove duplicates, handle missing values, fix formatting errors
Data Organization	Sort and filter large datasets
Data Analysis	Use formulas like SUM, AVERAGE, COUNT, IF, etc.
Data Exploration	Use Pivot Tables to find patterns and summarize data
Data Visualization	Create charts and graphs to understand trends
Reporting	Create reports and dashboards for decision-making

In Simple Words

Excel is used throughout the entire data analysis process — especially for data cleaning, analysis, visualization, and reporting.

- Fast and familiar — the spreadsheet interface most beginners already know
- Great for small to medium datasets and quick, ad-hoc analysis
- Builds the foundation for SQL queries and Power BI dashboards later on

Example: 10,000 Sales Records

Raw Data
↓
Clean Data
↓
Sort & Filter
↓
Pivot Table
↓
Charts
↓
Insights

Building My Excel Skillset

1 Beginner

- Data entry & formatting
- SUM, AVERAGE, COUNT
- Sorting & filtering
- Basic charts

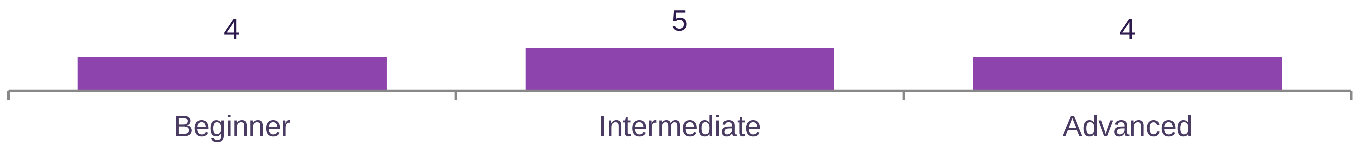
2 Intermediate

- IF & nested IF
- VLOOKUP / XLOOKUP
- Pivot Tables
- Conditional formatting
- SUMIFS / COUNTIFS

3 Advanced

- INDEX-MATCH & array formulas
- Power Query
- Power Pivot & DAX
- Macros / VBA

Skills Mapped by Level



Where to Spend My Time

Pivot Tables and VLOOKUP/XLOOKUP unlock the most value for the least effort — master those before moving on to Power Query or VBA.

Data Pipelines & the Data Engineer

What Is a Data Pipeline?

A process that moves data from one place to another and prepares it for use. A Data Engineer builds and manages these pipelines.

Customers place orders

↓

Data generated from website/app

↓

Pipeline collects the data

↓

Cleans & transforms the data

↓

Stores it in a data warehouse

↓

Data Analyst builds reports

E What a Data Engineer Does

- Collects data from different sources: databases, APIs, apps
- Cleans and transforms the data
- Moves data to a warehouse or data lake
- Automates the process so it runs regularly
- Monitors the pipeline and fixes errors

↔ Simple Difference

Data Engineer
Builds the pipeline that delivers reliable data.

Data Analyst
Uses that data to find insights and answer business questions.

Excel vs SQL vs Power BI

Point	Excel	SQL	Power BI
Main Purpose	Data entry, calculation, and analysis	Retrieving and managing data from databases	Data visualization and reporting
Data Size	Best for small to medium datasets	Handles very large datasets	Can visualize and analyze large datasets
Main Work	Formulas, Pivot Tables, data cleaning	Queries, filtering, joining tables	Dashboards, charts, interactive reports
Example	Calculate total sales using formulas	Find all customers who purchased a product	Create a dashboard showing monthly sales
Used By	Data Analysts, accountants, business users	Data Analysts, Data Engineers, developers	Data Analysts, BI professionals

X Excel — Best For

Quick analysis on small to medium datasets you already have in hand.

Q SQL — Best For

Pulling and shaping data directly from large databases.

P Power BI — Best For

Turning data into dashboards everyone on the team can explore.

Excel = analyze it yourself · SQL = pull it from the database · Power BI = show it to everyone else

Three Quick Comparisons

Excel vs SQL

Point	Excel	SQL
Type	Spreadsheet software	Query language
Main Purpose	Analyze & visualize data	Retrieve & manipulate data
Data Storage	Worksheets	Databases
Data Size	Small to medium	Very large
Common Tasks	Formulas, Pivot Tables, charts	SELECT, filter, join, aggregate

Excel vs Power BI

Point	Excel	Power BI
Main Purpose	Entry, calculation, analysis	Visualization & dashboards
Data Size	Small to medium	Large, handled efficiently
Visualization	Basic charts and graphs	Advanced interactive visuals
Data Refresh	Usually manual	Can auto-refresh
Best Use	Ad-hoc analysis & spreadsheets	Reporting & sharing insights

SQL vs Power BI

Point	SQL	Power BI
Main Purpose	Retrieve & manage data	Visualize data & dashboards
Type	Query language	BI tool
Main Work	Filter, join, update, aggregate	Charts, graphs, reports
Data Source	Works directly with databases	Connects to SQL, Excel, cloud
End Result	Returns data in table format	Interactive visual reports

EXCEL IN ONE WORD

Calculate

Turn raw numbers into totals, formulas, and quick summaries.

SQL IN ONE WORD

Retrieve

Pull exactly the records you need from a database.

POWER BI IN ONE WORD

Visualize

Turn queries and spreadsheets into shareable dashboards.

Excel & SQL: Excel is a spreadsheet, SQL is a language for talking to databases.

Together, they cover the full journey: pull the data, shape it, show it.

PUTTING IT TOGETHER

A Beginner's Learning Path

1
Excel
Clean, calculate, first insights

→

2
SQL
Query real databases

→

3
Power BI
Build shareable dashboards

→

4
Python
Automate & scale analysis

The Big Takeaway

Data Engineers build the pipeline. Data Analysts explore, calculate, and report on what flows through it — Excel for the first pass, SQL to reach the database directly, Power BI to share the story.

Different tools, same goal: turn raw data into a decision.